

EX NO.: 6	CHARACTERISTICS OF JUNCTION FIELD EFFECT TRANSISTOR.	12.09.17
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AIM:

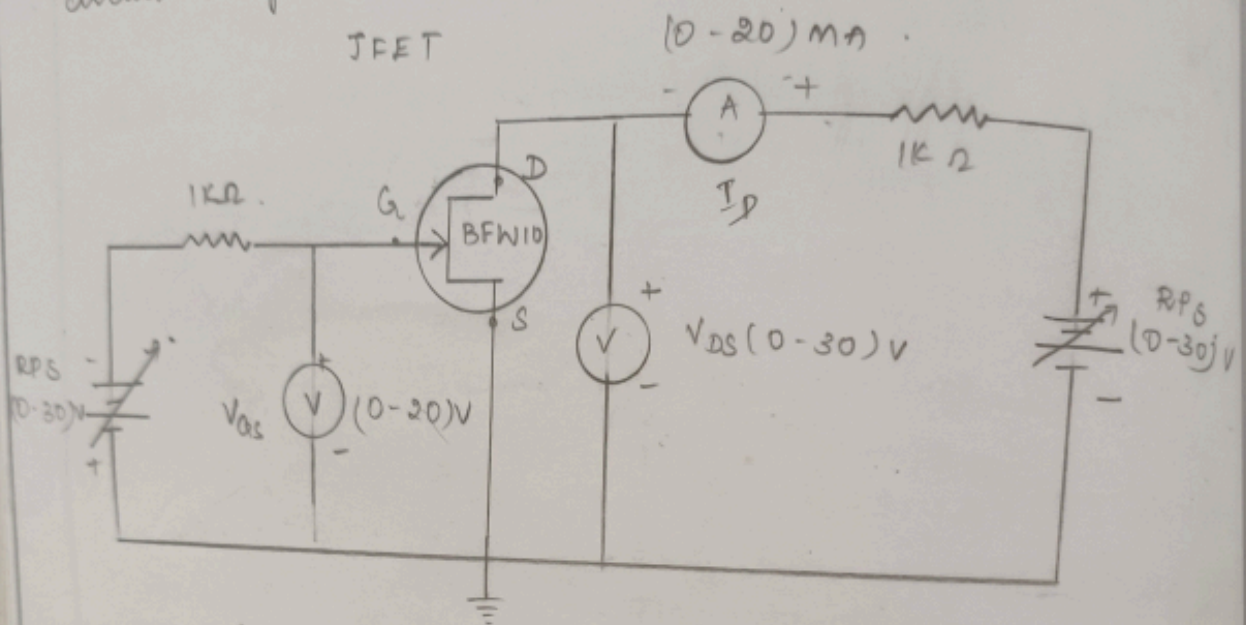
TO determine the output and transfer characteristics of field effect transistor and to determine the drain characteristics, trans conductance, amplification factor from the characteristics.

APPARATUS REQUIRED:

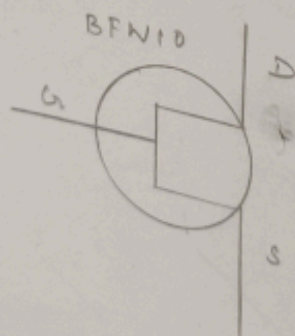
SL NO.	APPARATUS / COMPONENT NAME	TYPE / RANGE	QUALITY.
1.	Dual Regulated power supply (RPS)	(0-30)V	1
2.	Voltmeter	(0-10)V	1
3.	Ammeter	(0-20)MA	1
4.	Bread Board	-	1
5.	Junction field effect transistor (JFET)	BFW10	1
6.	point contact diode	0A79	1
7.	connecting wires	-	As required.

circuit diagram:

JFET

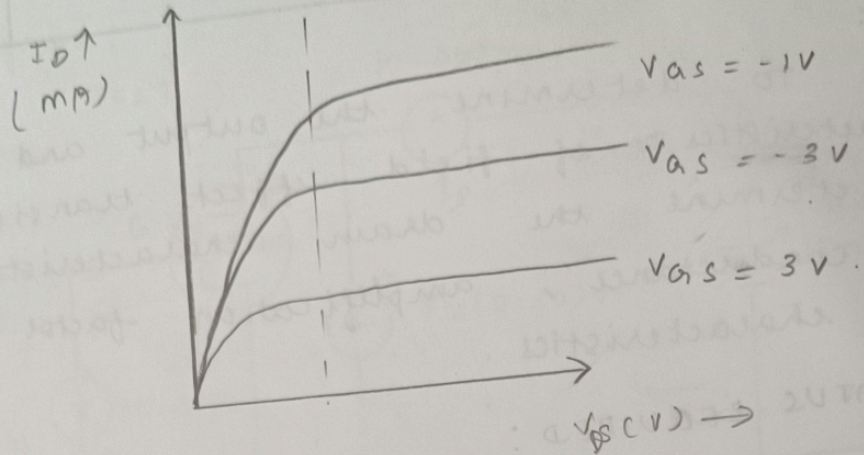


symbol



Model graph:

Active Region



JFET

TABULATION:

$V_{GS} = -1V$		$V_{GS} = -2V$		$V_{GS} = -3V$	
$V_{DS}(V)$	$I_D(mA)$	$V_{DS}(V)$	$I_D(mA)$	$V_{DS}(V)$	$I_D(mA)$
0.09	0.88	0.16	0.52	0.12	0.38
0.11	1.59	0.24	0.66	0.16	0.53
0.16	2.49	0.25	0.72	0.18	0.63
0.18	3.20	0.26	1.32	0.19	0.68
0.19	3.99	0.27	1.68	0.21	0.72
0.20	4.69	0.27	2.0	0.25	0.76
0.22	5.34	0.28	2.50	0.26	0.80
0.24	5.83	0.30	2.79	0.29	0.85
0.25	6.24	0.32	3.04	0.30	0.87
0.27	6.44	0.48	3.76	0.31	0.91
0.28	6.48	0.50	3.27	0.33	0.92
0.38	6.59	0.58	3.34	0.34	0.93
0.32	6.61	0.59	3.41	0.35	0.94
0.33	6.63	0.68	3.47	0.40	0.95
0.35	6.63	0.72			

FORMULA :

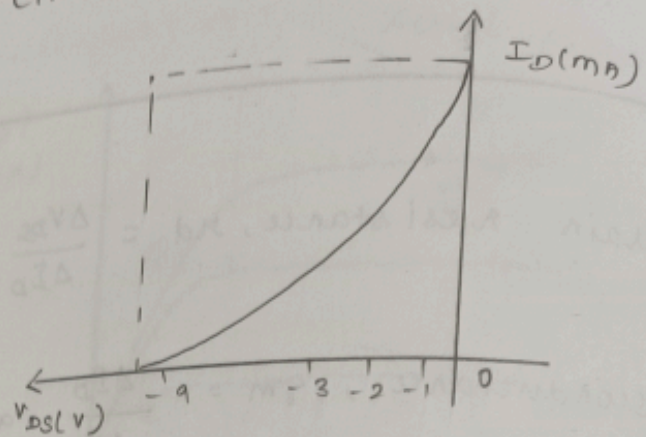
1. AC drain resistance, $r_d = \frac{\Delta V_{DS}}{\Delta I_D}$ at constant ΔV_{GS} .
2. Transconductance, $g_m = \frac{\Delta I_D}{\Delta V_{GS}}$ at constant ΔV_{DS} .
3. Amplification factor $\mu = r_d \times g_m$.

THEORY :

A FET (Field Effect transistor) is a three terminal device, in which current conduction is by majority carriers only. The flow of current is controlled by means of an electric field. The three terminals of FET are gate, Drain and Source. It is having the characteristic of high input impedance and less noise, the gate to source junction of the FETs always reverse biased. In response to small applied voltage from drain to source, the type bar act as sample resistor and the drain current increases linearly with V_{DS} with increase in I_D the ohmic voltage drop between the



TRANSFER CHARACTERISTICS



TABULATION:

$V_{DS} = 1V$		$V_{DS} = 3V$		$V_{DS} = 5V$	
$V_{GS}(V)$	$I_D(mA)$	$V_{GS}(V)$	$I_D(mA)$	$V_{GS}(V)$	$I_D(mA)$
0	4.97	0	7.68	0	8.6
-1.06	4.78	-1.01	6.84	-0.91	7.32
-1.33	4.59	-1.37	5.79	-1.32	6.28
-1.77	3.99	-1.45	3.39	-1.93	4.0
-2.42	2.26	-1.59	1.90	-2.54	2.11
-3.08	0.82	-2.69	0.98	-2.87	1.26
-3.54	0.04	-3.43	0	-3.53	0.1
-5.11	0	-4.13	0	-4.02	0
-4.47	0	-4.9	0	-4.55	0
-5.09	0			-5.11	

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source and the channel region reverse biases the junction and the conducting position of the channel begins to remain constant. The V_{DS} at this instant is called "Pinch off voltage".

If the gate to source voltage (V_{GS}) is applied in the direction to provide additional reverse bias, the pinch off voltage is decreased. In amplifier application, the FET is always used in the region beyond the pinch-off.

ACTIVE REGION:

In this region theory for a small increase in drain to source voltage the drain current increases largely. Hence this called active region.

CUT OFF REGION:

In this region the current never increases even if voltage increases. Hence it is called cut off region.

STATURATION REGION:

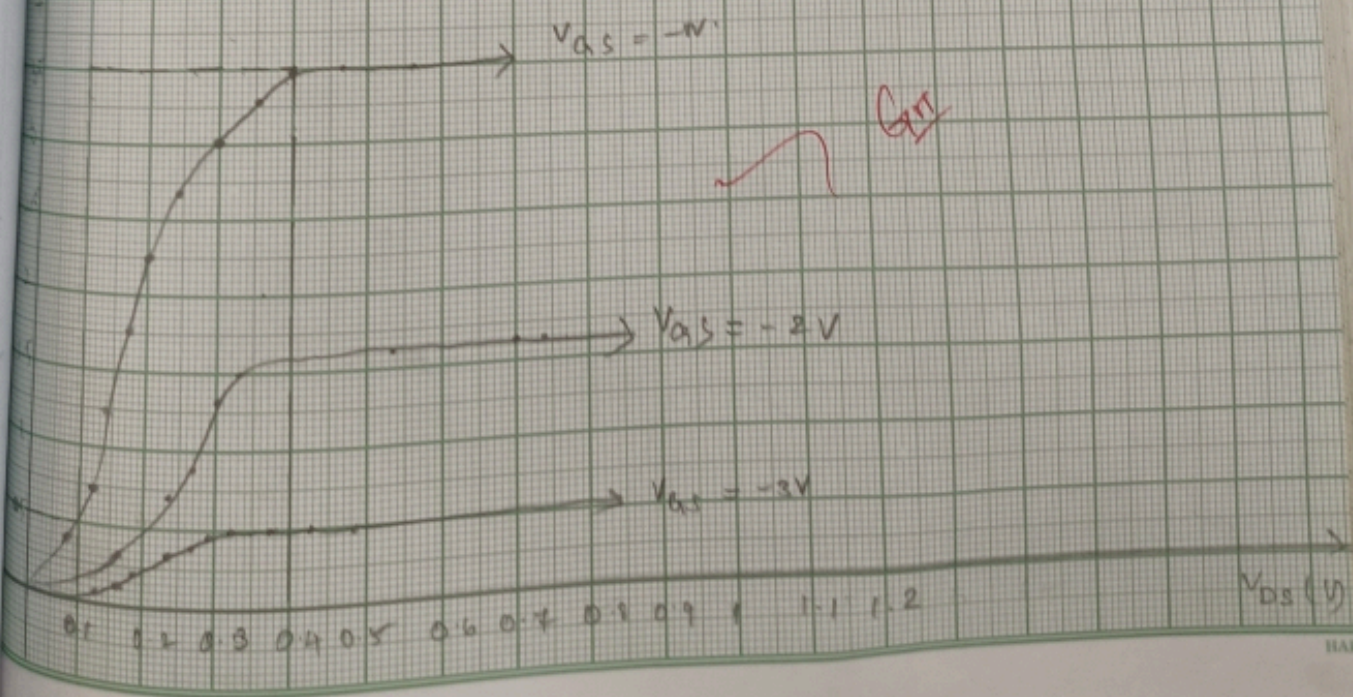
In this region after increasing from active region, the drain current remains constant over a range of drain



Characteristics of JFET

Drain characteristics

Scale:
On x axis 1 cm = 0.1 V
On y axis 1 cm = 0.5 mA



$$r_d = \frac{\Delta V_{DS}}{\Delta I_D}$$

$$= \frac{6.1 - 0.37}{1.2 - 6.4}$$

$$= 51.92 \times 10^{-3} \Omega$$

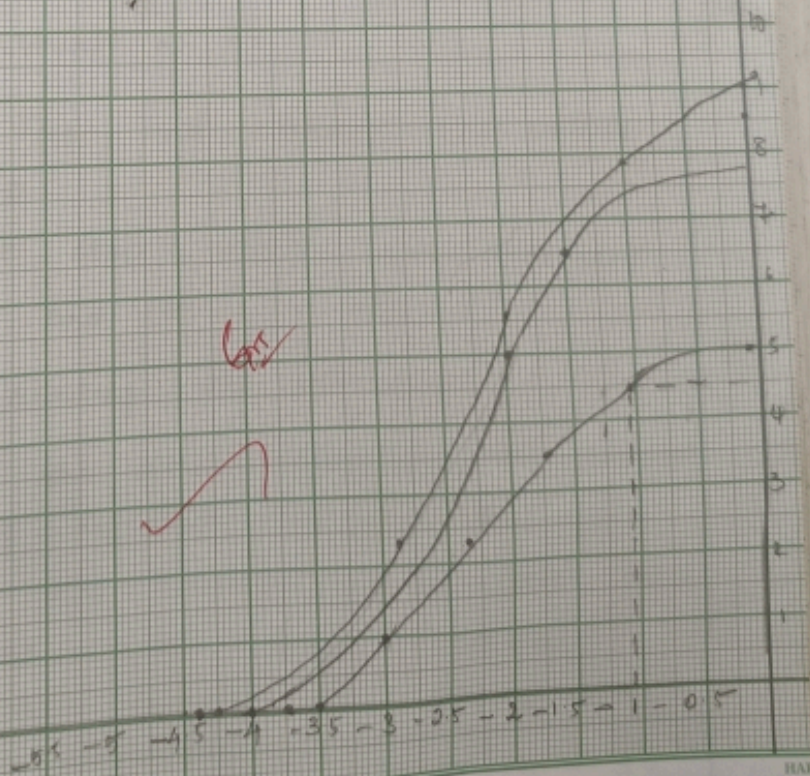
characteristics of JFET

I_{DS}

Scale

X axis 1 cm = 0.5 V

Y axis 1 cm = 1 mA



$$g_m = \frac{\Delta I_D}{\Delta V_{GS}}$$

$$= \frac{5 - 4.6}{1.2 - 0}$$

$$= 333.3 \times 10^{-3} \Omega$$

$$\text{Amplification factor } \mu = r_d \times g_m$$

$$= 51.92 \times 10^{-5} \times 333.3$$

$$= 17.30 \times 10^{-3} \Omega$$

to source voltage. After that current pinches off or shoots to a very high value.

PROCEDURE

- * All the connections are made as per the circuit diagram.
- * To plot the drain characteristics keep V_{GS} constant at 0V.
- * Vary the V_{DS} and observe the values of V_{DS} and I_D .
- * Repeat the above steps 2, 3 for different values of V_{GS} at 0.1V and 0.2V.
- * All the readings are tabulated.
- * To plot the transfer characteristics, keep V_{DS} constant at 1V.
- * Vary V_{GS} and observe the values of V_{GS} and I_D .
- * Repeat step 6 and 7 for different values of V_{DS} and at 1.5V and 2V.
- * The readings are tabulated.
- * From the drain characteristics, calculate the values of dynamic resistance (r_d).
- * From the transfer characteristics, calculate the value of transconductance (g_m).
- * And also calculate the amplification factor (μ).



RESULT :

Thus the output and transfer characteristics of Junction field transistor (JFET) are determined. The following parameters are determined from the graph.

→ Ac drain resistance, $r_d = 51.92 \times 10^3 \Omega$

→ Transconductance, $g_m = 333.3 \times 10^3 \Omega$

→ Amplification factor, $\mu = 17.30 \times 10^3 \Omega$

